Higher National Diploma in Civil Engineering

Second Year, Second Semester Examination - 2018

CE2215- Structural Analysis

1.

a. Consider the simply supported beam AB of span L carrying a UDL of w per unit length and its mid-point is C as shown in Figure 01. Determine the **deflection at mid span C**, the flexural rigidity of the beam is EI. Using energy theory. (10 Marks)

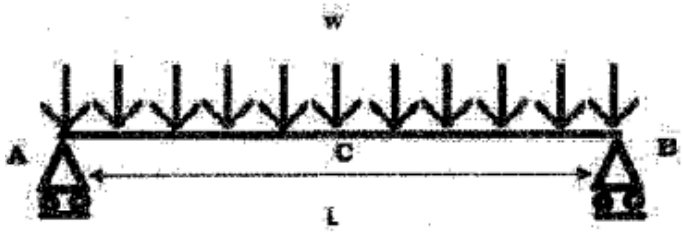

Figure 01

b. Find the **deflection at mid span C in mm** if UDL w = 20kN/m, and point load 10kN is applied at C. Length L = 5m. Young's modulus E = 210GPa second moment of inertia I = 1.8×10^5 m^4 using energy theory. (10 Marks)

2.

Analyze the beam ABC as shown in **Figure 02** by using **slope deflection method** and determine the **bending moment of the beam** also draw the bending moment diagram. The EI is constant throughout the beam. (20 Marks)

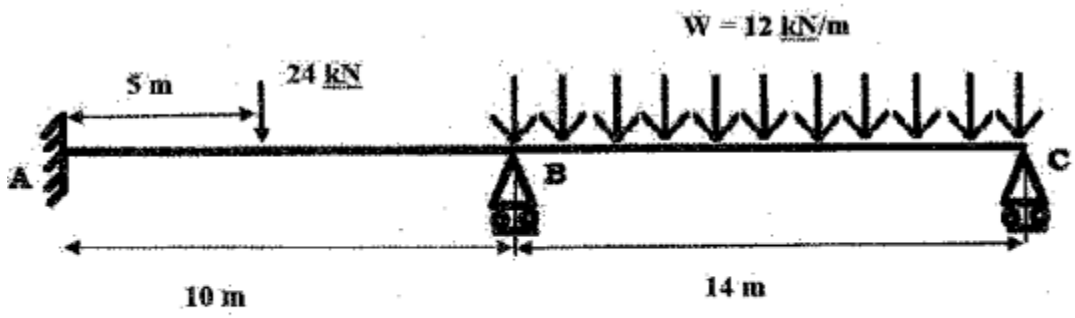

Figure 02

3.

A continuous beam ABC is subjected to uniformly distributed load 25kN/m and concentrated load of 60kN as shown in **Figure 03**. Flexural rigidity of the beam is EI.

a. Determine the moment distribution factor. (05 Marks)

b. Analyze the beam using moment distribution method and draw the bending moment diagram. (15 Marks)

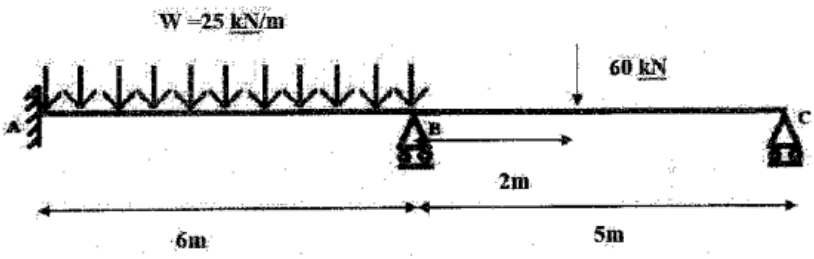

Figure 03

4.

a. Explain why the **method of sections is preferable** than a method of joints while analyzing the pin jointed trusses. (05 Marks)

b. Determine the **member forces** of the pin jointed truss using **method of joints** as shown in **Figure 04**. (15 Marks)

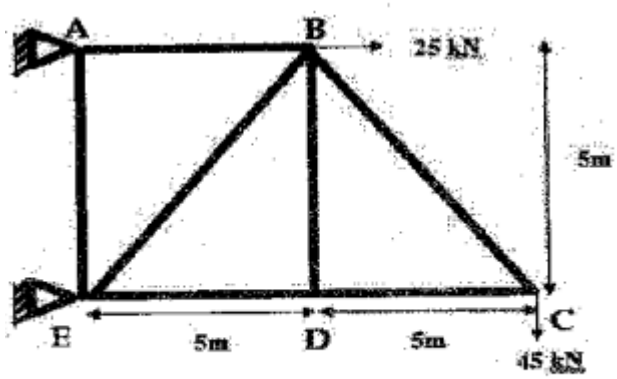

Figure 04

5.

A beam ABCD subjected to UDL and point load as shown in **Figure 05**. The modulus of elasticity (E) and second moment of inertia (I) are same for all members.

a. Determine the end moments using three moment method. (16 Marks)

b. Draw the Bending moment diagram. (04 Marks)

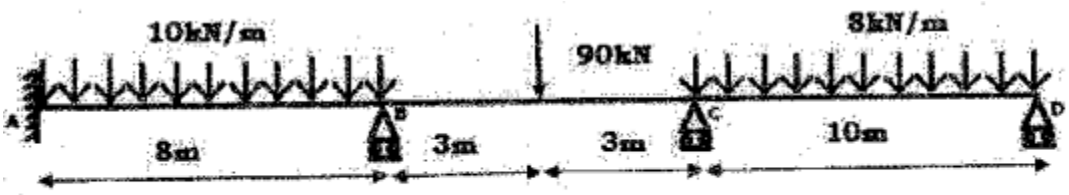

Figure 05

6.

Find the **horizontal reactions** of the **two hinged parabolic arch** as shown in **Figure 06** by neglecting shear and axial effect. (20 marks)

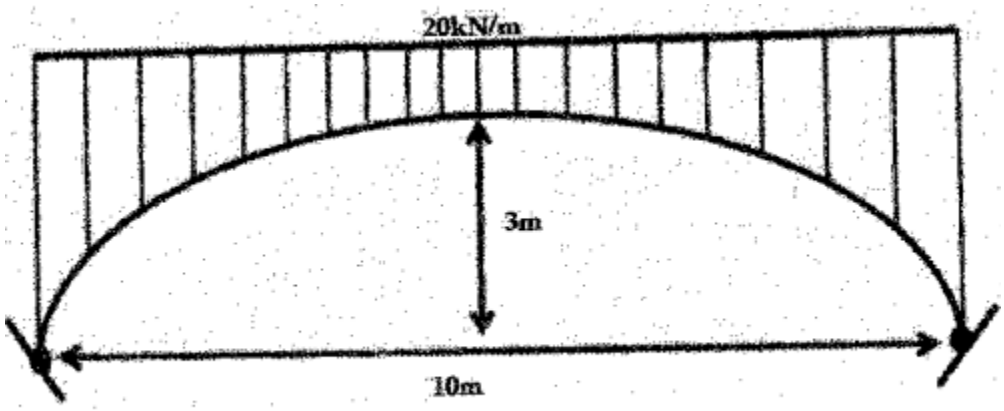

Figure 06